Automated Computer Vision Analysis of the Coefficient of Restitution of a Basketball Under Varying Inflation Pressures and Drop Heights

John Johnson

Department of Mechanical Engineering
Research under Haim Baruh
Rutgers University - New Brunswick

Abstract

This study investigated the effects of basketball inflation pressure and drop height on the coefficient of restitution (COR). A total of 90 trials were performed using three inflation pressures (3.5, 7.0, and 10.5 PSI) and three drop heights (0.5, 1.0, and 1.5 m). High-speed video recordings were analyzed using a custom MATLAB program, Rigid-Body Tracker V6-2, developed for this project by the author. The program automatically tracked the basketball, filtered unreliable feature points, and calculated the coefficient of restitution for the first three bounces of each trial. The results showed that increasing inflation pressure increased the coefficient of restitution, while increasing drop height produced a slight decrease. This project demonstrates the use of automated computer vision as an efficient tool for analyzing rigid-body motion and impact experiments.

1. Introduction

The coefficient of restitution (COR) describes how efficiently an object rebounds after a collision. For a basketball, variables such as inflation pressure and drop height can affect the amount of energy retained during impact. Understanding these effects is useful in sports engineering and impact mechanics.

This study investigated the effects of basketball inflation pressure and drop height on the coefficient of restitution using 90 controlled drop tests. High-speed video recordings were analyzed with a custom MATLAB program, Rigid-Body Tracker V6-2, developed by the author. The program automatically tracked the basketball, filtered unreliable feature points, and calculated the coefficient of restitution for the first three bounces of each trial.

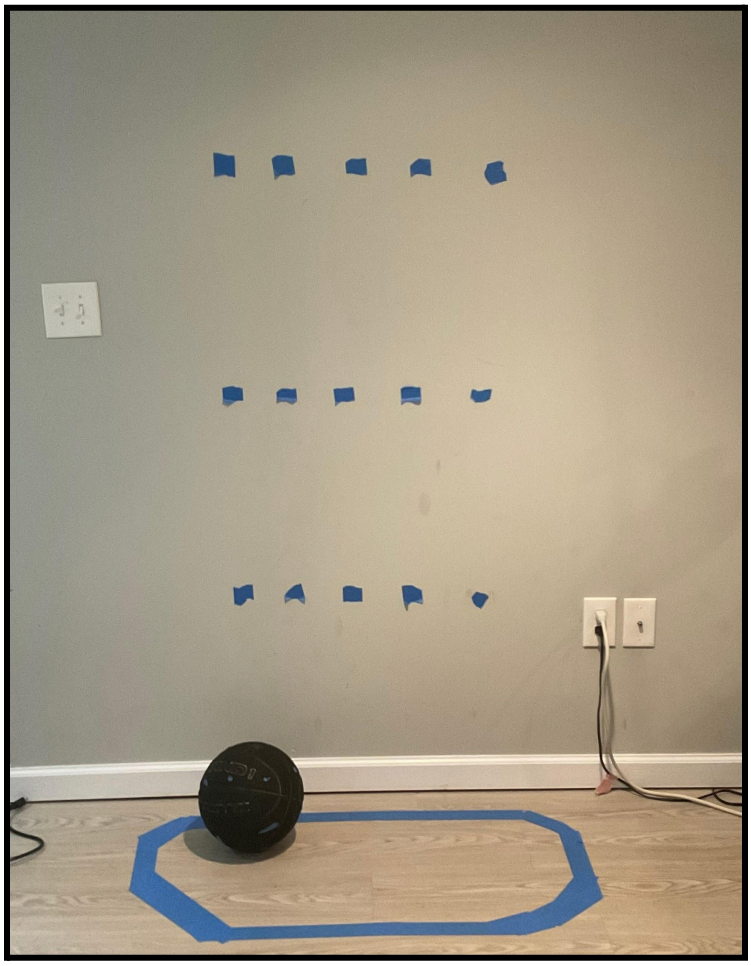
The objective of this study was to evaluate how inflation pressure and drop height influence basketball rebound behavior while demonstrating the use of automated computer vision for rigid-body motion analysis.

2. Experimental Methods

2.1 Experimental Design

Three inflation pressures (3.5, 7.0, and 10.5 PSI) and three drop heights (0.5, 1.0, and 1.5 m) were tested. Ten trials were performed for each pressure-height combination, resulting in a total of 90 trials. Each trial was recorded using a high-speed camera and analyzed after data collection.

Figure 1. Experimental setup showing the basketball, release height, and camera position. Each line of tape represents the 0.5m, 1.0m, and 1.5m heights.



2.2 Automated Motion Analysis

Video analysis was performed using Rigid-Body Tracker V6-2, a custom MATLAB program developed for this project by the author. The software automatically detects feature points on an object and tracks them throughout each trial.

Feature points with poor tracking quality, unpredictable movement, or not on the object are automatically filtered and removed before estimating the basketball's motion. The remaining feature points are used to determine the basketball's trajectory and calculate its kinematics throughout the experiment.

To evaluate the repeatability of the tracking program, several videos were analyzed multiple times using the same processing procedure. The calculated coefficients of restitution showed only small differences

between repeated analyses, indicating that the tracking algorithm produced consistent results when processing the same experimental data. This repeatability increased confidence that variations in the measured coefficients of restitution were due to the experimental conditions rather than inconsistencies in the analysis software.

Figure 2. Comparison of the coefficient of restitution obtained from two independent analyses of three randomly selected videos out of the 90.

Repeatability Verification of the Rigid-Body Tracker V6-2				
Video Index	Video	Δe1	Δe2	Δe3
#11	E3_3.5_1.0_V1	0.005	0.003	0.009
#45	E3_7.0_1.0_V5	0.004	0.006	0.004
#82	E3_10.5_1.5_V2	0.003	0.002	0.005
Average		0.004	0.004	0.006

The average repeatability differences (0.004–0.006) were considerably smaller than the average experimental standard deviation (~0.021), suggesting that software repeatability contributed relatively little to the overall variability of the measurements.

Figure 3. Example of kinematics data produced by the tracker.

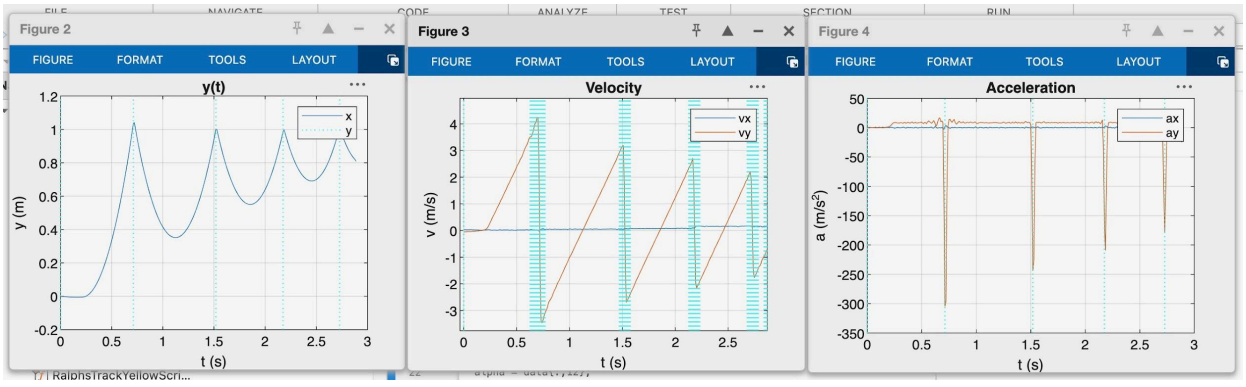
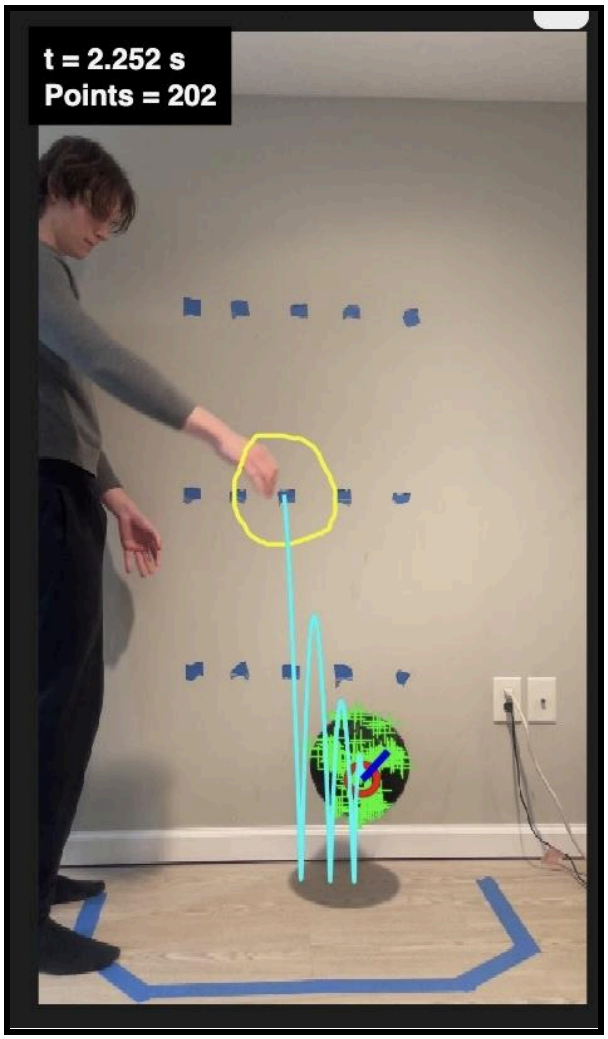


Figure 4. Example of detected feature points on the basketball.



3. Results

Increasing inflation pressure resulted in higher coefficients of restitution for all three bounces. The 10.5 PSI basketball consistently produced the highest rebound efficiency, while the 3.5 PSI basketball produced the lowest.

Increasing drop height produced a decreasing trend in the average coefficient of restitution. Although the trend was smaller than the effect of inflation pressure, it remained consistent across the tested conditions.

Figure 5. Average coefficient of restitution versus inflation pressure.

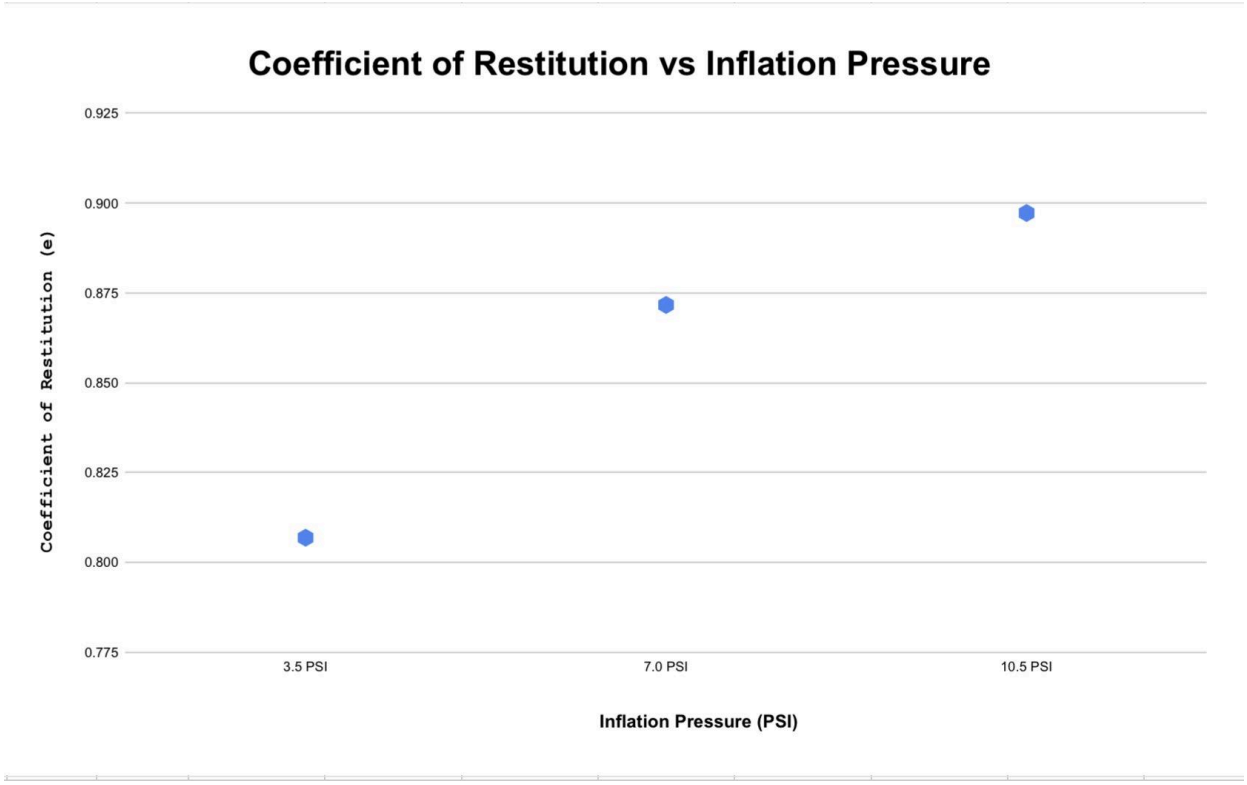
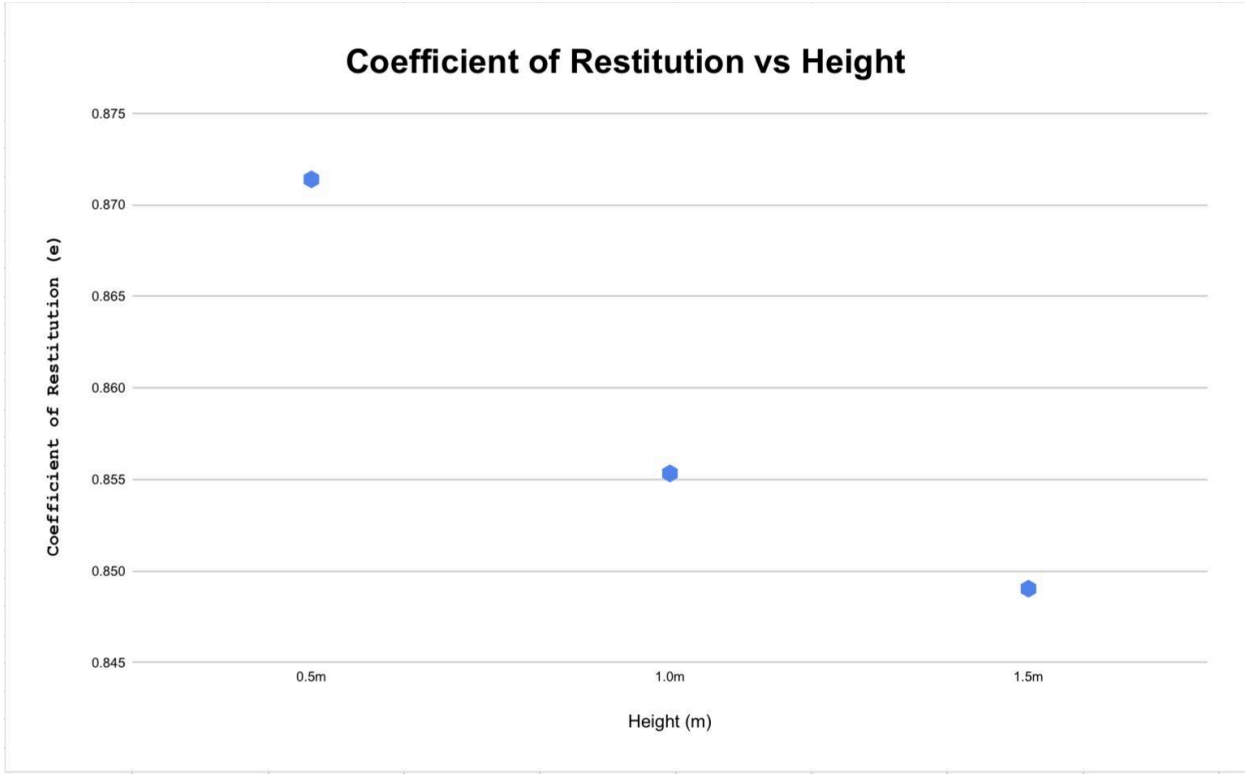


Figure 6. Average coefficient of restitution versus drop height.



Since a total of 90 trials were recorded, each datapoint in the graphs has 30 trials averaged.

In addition to the average coefficients of restitution, the standard deviation was calculated for each experimental condition. Across all pressure and height conditions, the average standard deviation was approximately 0.021, indicating relatively low trial-to-trial variability. This suggests that repeated trials produced consistent measurements and that the automated analysis provided repeatable results.

4. Discussion

The results indicate that both inflation pressure and drop height have an effect on the coefficient of restitution. A more highly inflated basketball deforms less during impact, allowing more energy to be returned during the rebound. This behavior agrees with the expectation that increased internal pressure allows the ball to return a greater fraction of its impact energy during rebound.

A ball dropped from higher up will have more velocity upon impact, which can lead to more energy loss upon the rebound.

The shape of both graphs suggests that the relationship may be nonlinear. Additional experimental conditions would be required to determine the underlying relationship.

The automated tracking program also proved effective for processing a large number of trials. By automating feature tracking and coefficient of restitution calculations, the software reduced manual analysis while providing a consistent method for every experiment. Repeated analysis of representative videos produced highly consistent coefficients of restitution (as seen from Figure 2), further demonstrating the repeatability of the analysis software.

Although the basketball temporarily deforms during impact, treating it as a rigid body provided accurate trajectory estimates for calculating rebound velocities. Future work could investigate tracking methods that account for ball deformation or evaluate additional basketballs and impact surfaces.

5. Conclusion

This study examined the effects of basketball inflation pressure and drop height on the coefficient of restitution using automated computer vision analysis. Higher inflation pressures produced higher coefficients of restitution, while greater drop heights produced a lower coefficient of restitution.

The custom MATLAB Rigid-Body Tracker V6-2 successfully automated motion tracking and coefficient of restitution calculations, demonstrating the usefulness of computer vision for experimental impact analysis.